

**Note → project on ec2 ubuntu machine , load balancing algorithms
→round robin , Leastconn & Health check using Apache2**

```
ubuntu@ip-172-31-4-250: /va  ×  +  v
ubuntu@ip-172-31-4-250:~$ cd /var/www/
ubuntu@ip-172-31-4-250:/var/www$ ls
dilip1  dilip2  dilip3  html
ubuntu@ip-172-31-4-250:/var/www$ cd dilip1
ubuntu@ip-172-31-4-250:/var/www/dilip1$ ls
index.html
ubuntu@ip-172-31-4-250:/var/www/dilip1$ |
```

```
ubuntu@ip-172-31-4-250: /etc  ×  +  v
ubuntu@ip-172-31-4-250:~$ cd /etc/apache2/
ubuntu@ip-172-31-4-250:/etc/apache2$ ls
apache2.conf  conf-available  conf-enabled  envvars  magic  mods-available  mods-enabled  ports.conf  sites-available  sites-enabled
ubuntu@ip-172-31-4-250:/etc/apache2$ cat ports.conf
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf
#Listen 80
Listen 8081
Listen 8082
Listen 8083

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
ubuntu@ip-172-31-4-250:/etc/apache2$ cd sites-available/
ubuntu@ip-172-31-4-250:/etc/apache2/sites-available$ ls
000-default.conf  default-ssl.conf  web1.conf  web2.conf  web3.conf
ubuntu@ip-172-31-4-250:/etc/apache2/sites-available$ |
```

```

ubuntu@ip-172-31-4-250:/etc/apache2$ cd sites-available/
ubuntu@ip-172-31-4-250:/etc/apache2/sites-available$ ls
000-default.conf  default-ssl.conf  web1.conf  web2.conf  web3.conf
ubuntu@ip-172-31-4-250:/etc/apache2/sites-available$ cd ..
ubuntu@ip-172-31-4-250:/etc/apache2$ cd ..
ubuntu@ip-172-31-4-250:/etc$ cd haproxy/
ubuntu@ip-172-31-4-250:/etc/haproxy$ ls
errors  haproxy.cfg
ubuntu@ip-172-31-4-250:/etc/haproxy$ cat haproxy.cfg
global
    log /dev/log local0
    log /dev/log local1 notice
    daemon
    maxconn 2048

defaults
    mode http
    log global
    option httplog
    timeout connect 5000ms
    timeout client 50000ms
    timeout server 50000ms

frontend http_front
    bind *:80
    default_backend web_cluster

backend web_cluster
    balance roundrobin
    option httpchk GET /
    http-check expect status 200

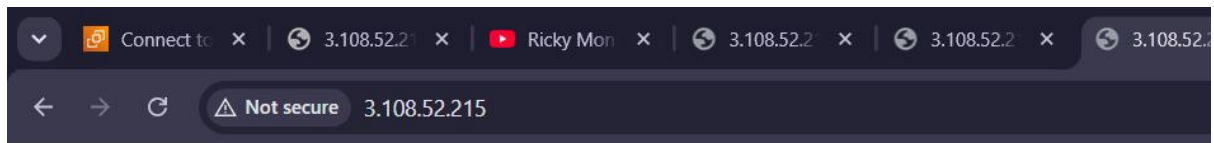
    server web1 127.0.0.1:8081 check inter 5s fall 3 rise 2
    server web2 127.0.0.1:8082 check inter 5s fall 3 rise 2
    server web3 127.0.0.1:8083 check inter 5s fall 3 rise 2
ubuntu@ip-172-31-4-250:/etc/haproxy$ |

```

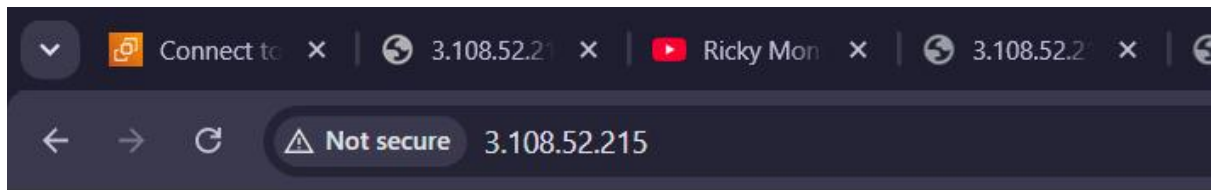
sudo systemctl restart haproxy

Sudo a2ensite filename → enable like it move the file into the sites enabled

port 80 I have to stop



Hello 1



Hello 2

← → ↻ ⚠ Not secure 3.108.52.215

Hello 3

PROJECT - ON EC2 Ubuntu Machine

Load Balancing Algorithms → Round Robin, Leastconn & Health Checks using Apache2

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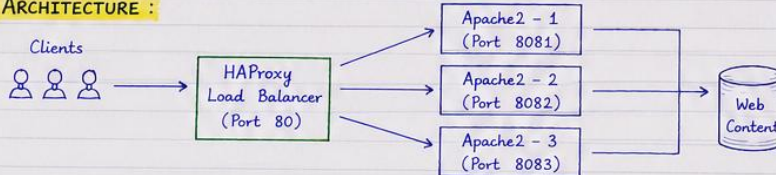


HAProxy + Apache2 = High Availability

OBJECTIVE :

Configure HAProxy as a load balancer in front of multiple Apache2 servers on a single EC2 Ubuntu machine and implement Round Robin, Leastconn algorithms with Health Checks.

ARCHITECTURE :



We Will Implement :

- Round Robin (default)
- Leastconn
- Health Checks

PROJECT STEPS :

1) UPDATE SYSTEM

```
sudo apt update
sudo apt upgrade -y
```

2) INSTALL APACHE2 (For Backend Servers)

```
sudo apt install apache2 -y
```

3) CREATE 3 APACHE2 INSTANCES (On Same EC2)

- We will run 3 Apache2 servers on different ports.
- Create 3 directories for content.

```
sudo mkdir -p /var/www/web1
sudo mkdir -p /var/www/web2
sudo mkdir -p /var/www/web3
```

- Create index.html for each:

```
/var/www/web1/index.html
<h1> Welcome to Web Server 1 </h1>
```

```
/var/www/web2/index.html
<h1> Welcome to Web Server 2 </h1>
```

```
/var/www/web3/index.html
<h1> Welcome to Web Server 3 </h1>
```

- Create 3 Apache2 Config Files:

```
sudo nano /etc/apache2/ports.conf ← Add Listen ports
```

```
Listen 8081
Listen 8082
Listen 8083
```

```
sudo nano /etc/apache2/sites-available/web1.conf
```

```
<VirtualHost *:8081>
    DocumentRoot /var/www/web1
    <Directory /var/www/web1>
        Require all granted
    </Directory>
    ErrorLog ${APACHE_LOG_DIR}/web1_error.log
    CustomLog ${APACHE_LOG_DIR}/web1_access.log combined
</VirtualHost>
```

Similarly create web2.conf (port 8082) and web3.conf (port 8083)

```
sudo a2ensite web1.conf web2.conf web3.conf
sudo systemctl restart apache2
```

4) VERIFY BACKEND APACHE2 SERVERS

- Open in browser or curl:
- `http://<EC2-IP>:8081` → Web Server 1
- `http://<EC2-IP>:8082` → Web Server 2
- `http://<EC2-IP>:8083` → Web Server 3

5) INSTALL & CONFIGURE HAPROXY

```
sudo apt install haproxy -y
```

Edit HAProxy Configuration:

```
sudo nano /etc/haproxy/haproxy.cfg
```

```
global
    log /dev/log local0
    log /dev/log local1 notice
    daemon
    maxconn 2048

defaults
    mode http
    log global
    option httplog
    timeout connect 5000ms
    timeout client 50000ms
    timeout server 50000ms

frontend http_front
    bind *:80
    default_backend web_cluster

backend web_cluster
    balance roundrobin # Change algorithm here
    option httpchk GET /
    http-check expect status 200
    server web1 127.0.0.1:8081 check inter 5s fall 3 rise 2
    server web2 127.0.0.1:8082 check inter 5s fall 3 rise 2
    server web3 127.0.0.1:8083 check inter 5s fall 3 rise 2
```

- * Change 'balance' to roundrobin or leastconn as per test.

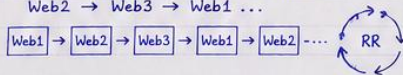
Restart HAProxy:

```
sudo systemctl restart haproxy
```

OK

6) TEST ROUND ROBIN (Default)

- Access `http://<EC2-Public-IP>/` multiple times.
- You will see requests going to Web1 → Web2 → Web3 → Web1 ...



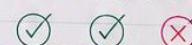
7) TEST LEASTCONN

- Change algorithm in backend:
- `balance leastconn`
- Restart haproxy
- Open multiple tabs (or use ab tool)
- Requests go to server with least active connections.



8) HEALTH CHECKS

- HAProxy checks backend servers using `httpchk (GET /)`.
- If a server fails (e.g., Apache stop), HAProxy will stop sending traffic to that server automatically.



SIMULATE FAILURE (Test Health Check)

- Stop one Apache server → `sudo systemctl stop apache2@8082`
- Refresh browser - traffic will not go to stopped server.
- Start again → `sudo systemctl start apache2@8082`
- It will rejoin automatically.



USEFUL COMMANDS

- Check HAProxy status : `sudo systemctl status haproxy`
- Check logs : `sudo tail -f /var/log/haproxy.log`
- Check Apache status : `sudo systemctl status apache2`

